

ZIAD MOHAMED EL-SHAHAT

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Objective

Robotics and Mechatronics Engineering student at Helwan National University with strong expertise in ROS 2, C++, Python, and embedded hardware development. Experienced in designing autonomous systems, microcontroller communication using ESP32 and Raspberry Pi 4, and wireless control protocols. Experienced in student engineering teams and robotics competitions, with a strong interest in robotics, automation, and embedded systems.

Education

September 2025 - September 2030

Bachelor of Mechatronics Engineering, Helwan National University.

Experience

Autonomous Landmine-Detection Robot | Minesweepers Competition (In Progress)

- Developed software nodes and navigation algorithms as part of a 10-member software engineering team.
- Configured ROS 2 (Jazzy) architecture on Ubuntu 24.04, including custom packages, launch files, and system parameters.
- Implemented ROS 2 node-to-node communication and integrated sensors for autonomous landmine detection.
- Contributed to the development, integration, and testing of the robot's autonomous software system.

Wireless Embedded Robotics Controller | ESP32 & Raspberry Pi 4 (June 2026)

- Designed and programmed a low-latency wireless control system using ESP32 and Raspberry Pi 4.
- Implemented wireless communication using ESP-NOW and hardware serial communication using UART.
- Integrated motor drivers and custom control logic for real-time robotic locomotion.
- Tested and debugged communication and control systems to ensure reliable robot operation.

Remote-Controlled (RC) Vehicle Project | MechaTech Team (June 2026)

- Co-founded MechaTech, a student engineering team focused on designing and developing robotic systems.
- Developed hardware wiring, power distribution, and microcontroller programming for an RC vehicle.
- Integrated electronic components, motor drivers, and control systems.

- Managed technical tasks, team workflow, and hardware assembly across the engineering team.

Training

Nassar Pipe and Plastic Factories | Industrial Engineering Trainee

Al-Obour City, Cairo, Egypt | 90 Days

- Completed hands-on training in an industrial manufacturing and plastic production facility.
- Observed industrial automation processes, manufacturing operations, and motor control systems.
- Examined motor control panels and industrial electrical systems.
- Applied engineering concepts to analyze real-world production and industrial environments.
- Learned workplace safety procedures and industrial engineering practices.

Certifications & Courses

- ROS 2 Development / ROS 2 from Scratch – Packt — August 2026
- AI for You – Machine Learning, Deep Learning & Generative AI Concepts — July 2026
- Python Programming Certification – Mahara-Tech / MCIT — July 2026
- Front-End Web Development – Ministry of Communications and Information Technology (MCIT) — May 2026

Skills

- Programming: C++, Python, Java, Bash, HTML5, CSS3.
- Robotics: ROS 2 (Jazzy), ROS 2 Nodes, ROS 2 Packages, Launch Files, Colcon, Autonomous Robotics.
- Operating Systems: Ubuntu 24.04, Linux.
- Embedded Systems: ESP32, Raspberry Pi 4, Microcontrollers, Motor Drivers.
- Communication Protocols: ESP-NOW, UART, Serial Communication.
- Engineering: Robotics Systems, Mechatronics, Automation, Control Systems, Hardware Integration.
- Soft Skills: Teamwork, Technical Leadership, Problem Solving, Communication, Task Management.
- Languages : Arabic (native language), English .